Linear Equations in One Variable

Concise Learning and Practice Guide

1. What does solving mean?

E1: A taxi charges a fixed fee of \$4 plus \$2 for each kilometre travelled. Suppose the total fare is \$18. **How many kilometres did the taxi travel?**

Let k = number of kilometres travelled. We can write $4 + 2k = 18$. To **solve** the equation means to find the value of k that makes the statement true:

$$4 + 2(7) = 18 \qquad \Rightarrow \qquad k = 7$$

The taxi travelled **7 km**.

E2: Two phone plans have different costs. Plan A charges \$10 plus \$3 per month, while Plan B charges \$22 plus \$2 per month. **After how many months will both plans have the same total cost?**

Let m = number of months. We can write $10 + 3m = 22 + 2m$. Solving gives

$$3m - 2m = 22 - 10 \qquad \Rightarrow \qquad m = 12$$

After **12 months**, both plans have the same total cost.

2. Basic solving process

Equations such as $ax + b = c$ and $ax + b = cx + d$ can be solved by:

1. Simplifying each side (distribute and combine like terms);
2. Collecting variable terms on one side;
3. Collecting constants on the other side;
4. Isolating the variable; and
5. Checking by substitution in the *original* equation.

Model: $3x + 5 = 20$

$$\begin{aligned} 3x + 5 - 5 &= 20 - 5 && \text{(subtract 5)} \\ 3x &= 15 \\ \frac{3x}{3} &= \frac{15}{3} && \text{(divide by 3)} \\ x &= 5 \end{aligned}$$

Check: $3(5) + 5 = 20$, so the left side equals the right side. Thus $x = 5$.

Model: $5x + 4 = 2x + 19$

$$\begin{aligned} 5x - 2x + 4 &= 2x - 2x + 19 && \text{(subtract } 2x\text{)} \\ 3x + 4 &= 19 \\ 3x + 4 - 4 &= 19 - 4 && \text{(subtract 4)} \\ 3x &= 15 \\ \frac{3x}{3} &= \frac{15}{3} && \text{(divide by 3)} \\ x &= 5 \end{aligned}$$

Check: $5(5) + 4 = 29$ and $2(5) + 19 = 29$, so both sides are equal. Thus $x = 5$.

3. Worked examples

A. Form: $ax + b = c$

$$\begin{aligned} -4x - 7 &= 17 \\ -4x - 7 + 7 &= 17 + 7 && \text{(add 7)} \\ -4x &= 24 \\ \frac{-4x}{-4} &= \frac{24}{-4} && \text{(divide by } -4\text{)} \\ x &= -6 \end{aligned}$$

Check: $-4(-6) - 7 = 17$.

B. Form: $ax + b = cx + d$

$$\begin{aligned} -5x + 4 &= 2x - 17 \\ -5x - 2x + 4 &= 2x - 2x - 17 && \text{(subtract } 2x\text{)} \\ -7x + 4 &= -17 \\ -7x + 4 - 4 &= -17 - 4 && \text{(subtract 4)} \\ -7x &= -21 \\ \frac{-7x}{-7} &= \frac{-21}{-7} && \text{(divide by } -7\text{)} \\ x &= 3 \end{aligned}$$

Check: $-5(3) + 4 = -11 = 2(3) - 17$.

C. Form: $a(x + b) = c$

$$\begin{aligned} -3(x - 2) &= 18 \\ -3x + 6 &= 18 && \text{(distribute } -3\text{)} \\ -3x + 6 - 6 &= 18 - 6 && \text{(subtract 6)} \\ -3x &= 12 \\ \frac{-3x}{-3} &= \frac{12}{-3} && \text{(divide by } -3\text{)} \\ x &= -4 \end{aligned}$$

Check: $-3(-4 - 2) = 18$.

D. Form: $\frac{x}{a} + b = c$

$$\begin{aligned} -\frac{x}{4} + 3 &= 8 \\ -\frac{x}{4} + 3 - 3 &= 8 - 3 && \text{(subtract 3)} \\ -\frac{x}{4} &= 5 \\ (-4)\left(-\frac{x}{4}\right) &= (-4)(5) && \text{(multiply by } -4\text{)} \\ x &= -20 \end{aligned}$$

Check: $-\frac{-20}{4} + 3 = 8$.

4. Equations in context

Follow the chain

$$\text{context} \longrightarrow \text{equation} \longrightarrow \text{solution} \longrightarrow \text{interpretation}.$$

Bike rental. A bike rental service charges a \$6 unlocking fee and \$3 per hour. The total cost is \$24. Let h be the number of hours.

$$6 + 3h = 24 \implies 3h = 18 \implies h = 6$$

The bike was rented for **6 hours**.

Online order. Four identical notebooks plus a \$3 delivery charge cost \$27. Let p be the price of one notebook.

$$4p + 3 = 27 \implies 4p = 24 \implies p = 6$$

Each notebook costs **\$6**.

Saving. Zuleika already has \$25 and saves \$8 each week. When will she have \$81? Let w be the number of weeks.

$$25 + 8w = 81 \implies 8w = 56 \implies w = 7$$

Zuleika needs **7 weeks**.

6. Practice problems

For Problems 1–6, **define a variable, construct an equation, solve, and interpret** the answer.

1. Two scooter companies offer different rental plans. Company A charges a \$6 unlocking fee plus \$4 per hour, while Company B charges a \$14 unlocking fee plus \$2 per hour. For how many hours is Company A cheaper than Company B?
2. Two streaming services offer different plans. Service A charges \$3 per premium movie with no monthly fee, while Service B charges a \$12 monthly fee plus \$1 per premium movie. For how many premium movies is Service A the cheaper option?
3. Cyprian is comparing two savings plans. Plan A starts with \$20 and adds \$8 each week, while Plan B starts with \$44 and adds \$5 each week. After how many weeks does Plan A contain more money than Plan B?
4. A taxi company charges \$5 per kilometre with no initial fee. Another transportation service charges a flat fee of \$30 regardless of the distance travelled. For what trip distances is the taxi company the cheaper option?
5. Two stores sell identical notebooks. Store A charges \$4 per notebook plus a \$10 delivery fee, while Store B charges \$6 per notebook with no delivery fee. For how many notebooks is Store A the cheaper option?
6. Two parking garages charge according to the length of a stay. Garage A charges \$3 per hour plus a \$5 entrance fee, while Garage B also charges \$3 per hour but has a \$9 entrance fee. For what lengths of stay is Garage A cheaper than Garage B? Explain what your result means in context.
7. Two bicycle rental shops use different pricing structures. Shop A charges \$4 per hour plus a \$6 rental fee. Shop B charges a \$2 processing fee, a \$4 service fee, and \$4 per hour. For what rental durations is Shop A cheaper than Shop B? Explain what your result means in context.

For Problems 8–25, solve each equation.

8. $x + 7 = 19$
9. $5y = 35$
10. $3z + 4 = 22$
11. $4a - 7 = 2a + 9$
12. $3(b + 2) = 21$
13. $2(c + 5) + 3 = 19$
14. $3(d + 2) + d = 18$
15. $2(m + 4) + m = 2m + 17$
16. $3(n + 2) + n = 2(n + 5) + 8$
17. $5(p - 3) + 2p = 3(p + 1) + 10$
18. $4(2q + 1) - 3q = 2(q + 7) + 5$
19. $7 - 2(r + 4) = 3(r - 1) - 5$
20. $3(2s - 5) - 2(s + 4) = 5 - (s + 6)$
21. $5 - 2\big(3v - (v - 4)\big) = 3(v + 1) - 4(2v - 1)$
22. $\dfrac{w + 5}{3} + \dfrac{w - 1}{2} = 7$
23. $\dfrac{h - 3}{4} - \dfrac{h + 1}{2} = \dfrac{h - 9}{8}$
24. $3(k - 2) + 2(2k + 1) = 7k - 4$
25. $4(2j - 3) - 2(j + 5) = 6j - 20$

7. Challenge problems

1. Find all real numbers x satisfying
$$3\big(2x - 5\big) - 2\big(3x - 7\big) = 5\big(x - 4\big) - \big(5x - 19\big).$$
Explain why the equation behaves differently from a typical linear equation.
2. Determine all real numbers x for which
$$\frac{2x - 3}{5} - \frac{3x + 1}{7} = \frac{4x - 9}{10} - \frac{6x + 5}{14}.$$
Do not begin by clearing all denominators. Look for a structural simplification first.
3. Let a be a real number. Consider the equation
$$(a + 1)\big((a - 1)x + 1\big) - 1 = (a + 1)x + 2.$$
Depending on the value of a, determine when the equation has:
   (a) exactly one real solution;
   (b) no solution;
   (c) infinitely many solutions.

   When the solution is unique, express x in terms of a.

8. Answer key

Practice problems

1. Let h be the number of rental hours.

$$6 + 4h < 14 + 2h$$

$$2h < 8 \quad \Longrightarrow \quad h < 4.$$

Therefore, Company A is cheaper for

$$\boxed{0 \le h < 4}\,.$$

2. Let m be the number of premium movies.

$$3m < 12 + m$$

$$2m < 12 \quad \Longrightarrow \quad m < 6.$$

Since m is a whole number, Service A is cheaper for

$$\boxed{m = 0, 1, 2, 3, 4, 5}\,.$$

3. Let w be the number of weeks.

$$20 + 8w > 44 + 5w$$

$$3w > 24 \quad \Longrightarrow \quad w > 8.$$

Therefore, Plan A contains more money

$$\boxed{\text{after more than 8 weeks}}\,,$$

or from week 9 onward.

4. Let d be the trip distance in kilometres.

$$5d < 30$$

$$d < 6.$$

Therefore, the taxi company is cheaper for

$$\boxed{0 \le d < 6 \text{ km}}\,.$$

5. Let n be the number of notebooks.

$$4n + 10 < 6n$$

$$10 < 2n \quad \Longrightarrow \quad n > 5.$$

Since n is a whole number, Store A is cheaper for

$$\boxed{n \ge 6}\,.$$

6. Let h be the number of parking hours.

$$3h + 5 < 3h + 9$$

$$5 < 9.$$

This statement is always true. Therefore, Garage A is cheaper for

$$\boxed{\text{every possible length of stay}}\,.$$

7. Let h be the rental duration in hours.

$$4h + 6 < 2 + 4 + 4h$$

$$4h + 6 < 4h + 6$$

$$6 < 6.$$

This statement is never true. Both shops charge exactly the same amount for every rental duration. Therefore,

$$\boxed{\text{Shop A is never cheaper than Shop B}}\,.$$

Equations

8.

$$x + 7 = 19 \quad \Longrightarrow \quad \boxed{x = 12}\,.$$

9.

$$5y = 35 \quad \Longrightarrow \quad \boxed{y = 7}\,.$$

10.

$$3z + 4 = 22 \quad \Longrightarrow \quad 3z = 18 \quad \Longrightarrow \quad \boxed{z = 6}\,.$$

11.

$$4a - 7 = 2a + 9$$

$$2a = 16 \quad \Longrightarrow \quad \boxed{a = 8}\,.$$

12.

$$3(b + 2) = 21$$

$$b + 2 = 7 \quad \Longrightarrow \quad \boxed{b = 5}\,.$$

13.

$$2(c + 5) + 3 = 19$$

$$2c + 13 = 19 \quad \Longrightarrow \quad 2c = 6 \quad \Longrightarrow \quad \boxed{c = 3}\,.$$

14.

$$3(d + 2) + d = 18$$

$$4d + 6 = 18 \quad \Longrightarrow \quad 4d = 12 \quad \Longrightarrow \quad \boxed{d = 3}\,.$$

15.

$$2(m + 4) + m = 2m + 17$$

$$3m + 8 = 2m + 17 \quad \Longrightarrow \quad \boxed{m = 9}\,.$$

16.

$$3(n + 2) + n = 2(n + 5) + 8$$

$$4n + 6 = 2n + 18$$

$$2n = 12 \quad \Longrightarrow \quad \boxed{n = 6}\,.$$

17.

$$5(p-3)+2p=3(p+1)+10$$
$$7p-15=3p+13$$
$$4p=28 \quad \Longrightarrow \quad \boxed{p=7}.$$

18.

$$4(2q+1)-3q=2(q+7)+5$$
$$5q+4=2q+19$$
$$3q=15 \quad \Longrightarrow \quad \boxed{q=5}.$$

19.

$$7-2(r+4)=3(r-1)-5$$
$$-2r-1=3r-8$$
$$7=5r \quad \Longrightarrow \quad \boxed{r=\frac{7}{5}}.$$

20.

$$3(2s-5)-2(s+4)=5-(s+6)$$
$$4s-23=-s-1$$
$$5s=22 \quad \Longrightarrow \quad \boxed{s=\frac{22}{5}}.$$

21.

$$5-2\big(3v-(v-4)\big)=3(v+1)-4(2v-1)$$
$$5-2(2v+4)=3v+3-8v+4$$
$$-4v-3=-5v+7 \quad \Longrightarrow \quad \boxed{v=10}.$$

22.

$$\frac{w+5}{3}+\frac{w-1}{2}=7$$

Multiplying by 6,

$$2(w+5)+3(w-1)=42$$
$$5w+7=42 \quad \Longrightarrow \quad \boxed{w=7}.$$

23.

$$\frac{h-3}{4}-\frac{h+1}{2}=\frac{h-9}{8}$$

Multiplying by 8,

$$2(h-3)-4(h+1)=h-9$$
$$-2h-10=h-9$$
$$-1=3h \quad \Longrightarrow \quad \boxed{h=-\frac{1}{3}}.$$

24.

$$3(k-2)+2(2k+1)=7k-4$$
$$7k-4=7k-4.$$

The equation is true for every real value of k. Therefore,

$$\boxed{k\in\mathbb{R}}.$$

25.

$$4(2j-3)-2(j+5)=6j-20$$
$$6j-22=6j-20$$
$$-22=-20.$$

This is false. Therefore,

$$\boxed{\text{no solution}}$$

Challenge problems – Answer key

1.

$$3(2x-5)-2(3x-7)=5(x-4)-(5x-19).$$

Simplifying each side,

$$6x-15-6x+14=5x-20-5x+19,$$

so

$$-1=-1.$$

This identity is true for every real number. Hence,

$$\boxed{x\in\mathbb{R}}.$$

The equation behaves differently from a typical linear equation because all terms containing x cancel and the remaining statement is always true.

2. Let

$$A=\frac{2x-3}{5}, \qquad B=\frac{3x+1}{7}.$$

Notice that

$$\frac{4x-9}{10}=\frac{2x-3}{5}-\frac{3}{10}=A-\frac{3}{10},$$

and

$$\frac{6x+5}{14}=\frac{3x+1}{7}+\frac{3}{14}=B+\frac{3}{14}.$$

Therefore, the equation becomes

$$A-B=A-\frac{3}{10}-\left(B+\frac{3}{14}\right).$$

Thus,

$$A-B=A-B-\frac{3}{10}-\frac{3}{14}=A-B-\frac{18}{35}.$$

Subtracting $A-B$ from both sides gives

$$0=-\frac{18}{35},$$

which is impossible. Therefore,

$$\boxed{\text{no real solution}}.$$

3. Consider

$$(a+1)\big((a-1)x+1\big)-1=(a+1)x+2.$$

Expand the left-hand side:

$$(a+1)(a-1)x+(a+1)-1=(a+1)x+2,$$

so

$$(a^2-1)x+a=(a+1)x+2.$$

Move all terms to one side:

$$(a^2-a-2)x+a-2=0.$$

Factor:

$$(a-2)(a+1)x+(a-2)=0,$$

and therefore

$$(a-2)\big((a+1)x+1\big)=0.$$

We now consider the special values of a.

- If $a=2$, then
$$0=0,$$
so the equation is true for every real x. Therefore,
$$\boxed{\text{infinitely many solutions}}.$$
- If $a=-1$, substituting into the original equation gives
$$-1=2,$$
which is impossible. Therefore,
$$\boxed{\text{no solution}}.$$
- If $a\neq 2$ and $a\neq -1$, we may divide by $a-2$:
$$(a+1)x+1=0.$$
Hence
$$x=-\frac{1}{a+1}.$$
Therefore, there is exactly one real solution:
$$\boxed{x=-\frac{1}{a+1}}.$$

Thus,

$$\boxed{\begin{cases}\text{infinitely many solutions}, & a=2,\\ \text{no solution}, & a=-1,\\ x=-\dfrac{1}{a+1}, & a\neq -1,2.\end{cases}}$$

9. Summarized solutions

Practice problems

1. $0 \le h < 4$
2. $m = 0, 1, 2, 3, 4, 5$
3. $w > 8$, so Plan A has more money from week 9 onward.
4. $0 \le d < 6$ km
5. $n \ge 6$
6. Garage A is cheaper for every possible length of stay
7. Shop A is never cheaper; both shops always charge the same

Equations

8. $x = 12$
9. $y = 7$
10. $z = 6$
11. $a = 8$
12. $b = 5$
13. $c = 3$
14. $d = 3$
15. $m = 9$
16. $n = 6$
17. $p = 7$
18. $q = 5$
19. $r = \frac{7}{5}$
20. $s = \frac{22}{5}$
21. $v = 10$
22. $w = 7$
23. $h = -\frac{1}{3}$
24. $k \in \mathbb{R}$
25. No solution

Challenge problems

1. $x \in \mathbb{R}$
2. No real solution
3. $$\begin{cases} \text{infinitely many solutions}, & a = 2, \\ \text{no solution}, & a = -1, \\ x = -\dfrac{1}{a+1}, & a \neq -1, 2. \end{cases}$$